EKF示例

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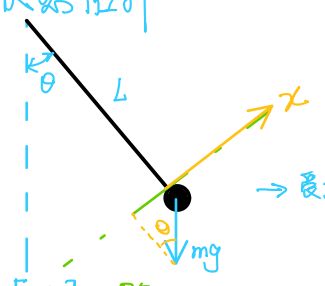
e.g. 单摆状态估计



→ 受力分析：切线方向：$-mg\sin\theta = mL\ddot{\theta}$

state： $x = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$ θ → 摆角，$\dot{\theta}$ → 角速度

1) 非线性状态方程（连续）

$$\begin{cases} \dot{\theta} = \dot{\theta} \\ \ddot{\theta} = -\frac{g}{L}\sin\theta \end{cases}$$

$x_1 = \theta,\ x_2 = \dot{x}_1 = \dot{\theta}$ $\Rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -\frac{L}{g}\sin x_1 \end{bmatrix} = f_c(x(t))$

2) 离散化状态方程

$$x_k = f(x_{k-1}) = x_{k-1} + \begin{bmatrix} \dot{\theta}_{k-1}\Delta t \\ -\frac{g}{L}\sin(\theta_{k-1})\Delta t \end{bmatrix} + w_{k-1} \Rightarrow f(x) = \begin{bmatrix} \theta + \dot{\theta}\Delta t \\ \dot{\theta} - \frac{g}{L}\sin\theta\Delta t \end{bmatrix} \begin{matrix} \to f_1 \\ \to f_2 \end{matrix}$$

3) 观测方程（非线性） 假设只能测量摆锤水平位置

$z_k = h(x_k) + v_k = L\sin\theta_k + v_k$ → 非线性观测

4) 雅可比矩阵

① State transition Jacobian F： $(x_1 = \theta,\ x_2 = \dot{\theta})$

$$F = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 1 & \Delta t \\ -\frac{g}{L}\cos\theta\Delta t & 1 \end{bmatrix}$$ 线性化点 $\hat{x}_{k-1}$

② Observation Jacobian H $H = \frac{\partial h}{\partial x}$

$$H = \begin{bmatrix} \frac{\partial h}{\partial \theta} & \frac{\partial h}{\partial \dot{\theta}} \end{bmatrix} = \begin{bmatrix} L\cos\theta & 0 \end{bmatrix}$$ 线性化点 $\hat{x}_k^-$

5) EKF 完整步骤

① 初始化： $\hat{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$，$P_0 = I$

② Predict

a. 非线性状态预测 $\hat{x}_k^- = f(\hat{x}_{k-1})$

b. Jacobian F（在 $\hat{x}_{k-1}$）： $F = \left.\frac{\partial f}{\partial x}\right|_{\hat{x}_{k-1}}$

c. 协方差预测： $P_k^- = FP_{k-1}F^T + Q$

③ Update

a. 非线性观测预测 $\hat{z}_k^- = h(x_k^-) = L\sin(\hat{\theta}_k^-)$

b. Jacobian $H(\hat{x}_k^-)$： $H = \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_k^-}$

c. Kalman Gain： $K_k = P_k^- H^T (HP_k^- H^T + R)^{-1}$

d. State update： $\hat{x}_k = \hat{x}_k^- + K_k(z_k - h(\hat{x}_k^-))$

e. Covariance update： $P_k = (I - K_k H)P_k^-$